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Rolled-sonicating enabled orientation of microstructures in flexible nanocomposites

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Abstract

Polymer nanocomposites filled with highly thermally conductive nano/microparticles have shown promise for efficient thermal management in power electronics due to their excellent flexibility and thermal conductivity. However, it remains challenging to align particulate fillers into an optimal heat transfer path through a fast and large-scale green fabrication process, especially when adapting to various types of fillers and flexible matrices. Herein, we report a unique rolled sonication-enabled microstructure orientation (RSMO) method that enables superfast densification and orientation of various fillers in polymers through the synergistic thermo-mechanical effects of high-frequency vibration with roller pressing, achieving full processing within one second without external heating. Ultrasound-induced dense packing and oriented structures synergistically enhance both thermal conductivity (an increase of 171%) and tensile strength (19-fold enhancement) compared with simple roller pressing. RSMO has been demonstrated to be applicable to a wide range of fillers, including metals, inorganic nonmetals, and organic materials, as well as various film-forming techniques, such as electrospinning and screen printing. We successfully fabricated large-area thermally conductive films using RSMO and demonstrated their thermal management capabilities in power electronics and flexible thermoelectric generation, thereby confirming their significant potential in preparing high-performance thermal management materials for flexible electronic devices.

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Keywords: sonicating process, large-scale fabrication, polymer nanocomposites, flexible electronics, thermal management

1. Introduction

The rapid development of high-power density integrated electronics has driven transformative progress in flexible electronics, such as foldable displays^[1–3], wearable bioelectronics^[4,5], soft robotics^[6], and sensors for extreme environments^[7–10]. However, inevitable heat accumulation during their operation not only significantly impairs working life and performance stability but also poses a fire risk. Thus, there is an urgent need for high-thermal-conductivity materials to enable efficient heat dissipation^[11]. Meanwhile, the unique deformation requirements of flexible electronic devices impose higher standards on the flexibility and fatigue resistance of heat-dissipating materials^[12,13]. For industrial applications, a stable, cost-effective production process is equally critical, as it directly corresponds to compatibility with large-scale manufacturing. In this context, these three core requirements—high thermal performance, excellent flexibility, and adaptability to mass manufacturing—drive researchers to advance thermal management materials and fabrication technologies that can effectively integrate all three characteristics.

At present, polymer-based thermally conductive nanocomposites have emerged as the mainstream research focus owing to their unique integration of the inherent flexibility of polymer matrices and the high thermal conductivity of thermally conductive fillers^[14–16]. The key to enhancing the thermal conductivity of such nanocomposites resides in constructing a continuous thermally conductive network. Specifically, modulating the microscopic orientation of thermally conductive fillers within the matrix reduces phonon interface scattering during propagation, which in turn minimizes thermal resistance^[17]. In existing research, widely utilized fillers include ceramics (Al_2O_3 , AlN , and boron nitride nanosheets (BNNs)), metals (Cu and Al), and carbon materials (graphene and CNTs)^[18]. However, within flexible thermally conductive polymer systems, the challenge of integrating rigid fillers with high thermal conductivity into flexible polymer matrices with low thermal conductivity persists^[11,19,20]. The simultaneous achievement of filler alignment and material flexibility remains a critical technical hurdle^[21–23].

Against this backdrop, researchers have devised a multitude of orientation regulation techniques, including directional freezing^[24,25], vacuum filtration^[26,27], blade coating^[28,29], electrospinning^[30,31], magnetic/electric field alignment^[32,33], hybrid filler systems^[34,35] and interfacial modifications^[36]. Nonetheless, a unidimensional approach often yields suboptimal filler orientation and is frequently associated with the development of detrimental mechani-

cal flaws, which inherently limit the overall performance of the nanocomposites. Therefore, numerous research groups have explored the combination of multiple technologies, such as orientation techniques coupled with pressing processes, which involve initial alignment and subsequent densification. The freezing orientation utilizes ice crystal growth to effectively promote vertical filler alignment. This treatment requires an ultra-low temperature environment ranging from 193.15 K to 233.15 K, and the duration can be as long as 12–72 hours^[37–39]. Furthermore, the freeze-drying process, which involves solvent evaporation, tends to leave numerous residual voids that induce substantial air thermal resistance, structural defects, and low mechanical strength, thereby reducing the material's heat transfer efficiency and bending resistance^[40]. Therefore, post-processing steps such as encapsulation or hot pressing are required before application^[11,24]. Its large-scale cost is extremely high because of prolonged ultra-low temperature treatment and custom large-scale equipment. Vacuum filtration facilitates the formation of highly aligned planar structures through gravity-driven sedimentation and negative-pressure consolidation^[41]. The single preparation process is limited by the filtration speed and takes 10–30 minutes, allowing the production of membranes with diameters of only 5–10 cm. When scaling up to larger areas ($>20 \text{ cm}^2$), maintaining a uniform pressure field becomes challenging. The resulting uneven negative pressure distribution on the membrane induces different sedimentation rates of the fillers. Consequently, this leads to structural heterogeneity, which significantly deteriorates the spatial consistency and reliability of the thermal conductivity of the final film^[42]. Moreover, the interfacial bonding between the polymer matrix and fillers remains insufficient following vacuum filtration, necessitating hot-pressing treatment to mitigate filler delamination or shedding^[27]. Blade coating utilizes shear forces to induce the ordered alignment of two-dimensional fillers within the polymer matrix^[29]. For the blade coating process, inevitable solvent evaporation during heating often leads to severe matrix distortion and concentration of internal stresses. These structural defects substantially compromise the mechanical reliability of the nanocomposites under tensile and bending loads^[43]. Similarly, in the fabrication of nanocomposites through electrospinning, fillers achieve a certain degree of in-plane orientation with significant inter-filler voids, which act as obstacles to phonon transport. Following this, a hot-pressing step is necessary to eliminate air gaps and concurrently regulate the compactness and increase the filler orientation. Studies have demonstrated that extended high-temperature pressing can significantly increase the thermal conductivity, glass transition temperature, and heat resistance index of polyvinyl alcohol (PVA)

nanocomposites^[14,44]. While hot-pressing has emerged as a conventional yet time-intensive technique for fabricating thermally conductive polymers, harsh processing conditions can lead to cracking or embrittlement in the film, thus impairing the flexibility of the composites^[45]. The current orientation strategies struggle to efficiently synergize mechanical properties and thermal conductivities in large-scale production. Thus, developing a multi-technology coupled approach, capable of precisely aligning fillers, enabling large-scale densification, and preserving the polymer matrix's flexibility, is essential for overcoming bottlenecks in flexible thermal management materials.

Herein, we report an innovative rolled sonicating-enabled microstructure-orientation (RSMO) method, which functions through the synergistic effect of high-frequency ultrasonic vibration and mild rolling pressure, facilitating the controlled alignment and densification of thermally conductive fillers in polymer matrices under low-pressure conditions without external heating. This transformation occurs within the span of a single second. This versatile approach is compatible with diverse film-forming methods (electrospinning or screen printing) and various filler types (metals, inorganic non-metals, organic materials, etc). By facilitating the horizontal alignment of fillers, the RSMO constructs continuous in-plane thermal conduction networks while minimizing matrix damage and porosity—synergistically enhancing both thermal conductivity and mechanical flexibility. For a PVA-based system as the testing model, the RSMO can achieve in-plane thermal conductivities of $11.54 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ with 30 wt% BNNSs and $7.53 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ with 30 wt% gallium (Ga)-based liquid metal, as well as high flexibility (bending radius $<1 \text{ mm}$), mechanical strength (tensile strength $>30 \text{ MPa}$), and over 90% thermal conductivity retention after repeated bending cycles. Notably, this method enables simultaneous cross-material coupling and orientation of heterogeneous films while also boasting potential for large-scale production. In addition, the effects of the process parameters on the degree of orientation and the thermal and mechanical properties of the nanocomposites are discussed. In real applications, RSMO-prepared flexible thermally conductive films demonstrate exceptional thermal management performance in light-emitting diode (LED) devices and flexible thermoelectric systems. This work provides new insights into the efficient preparation of flexible thermal management membranes and directional property regulation in multicomponent composites, offering potential solutions for high-power chips and flexible electronics in thermal management.

2. Results and discussion

2.1. RSMO-enabled composites densification and orientation

To intuitively illustrate the concept and principles of RSMO, electrospun films incorporating highly thermally conductive nano/micro-particles are used as a model system to demonstrate RSMO-enabled filler orientation and densification, as shown in Figure 1(a) (also see Figure S1 and Movie S1).

Under ultrasonic excitation, the sonotrode undergoes longitudinal high-frequency vibration, inducing periodic contact-separation behavior between the film to be processed and the rotating roller. This dynamic interaction enables the film to overcome static friction and continuously roll along the tangential direction of the drum (Figure 1(b)). Crucially, the synergy between ultrasonic vibration and mechanical rolling governs the internal structural evolution of the fillers. As depicted in Figure 1(a-ii), high-frequency ultrasound generates intense interfacial friction within the particle-embedded polymer matrix. The resulting thermal softening facilitates compatible plastic deformation^[50], ultimately yielding a densified film with minimized void volume. Simultaneously, under the coupled rolling-ultrasonic field (Figure 1(c)), the randomly distributed fillers are subjected to plastic flow, stress superposition, and acoustic streaming forces. These combined effects drive the fillers to rotate and align horizontally^[51–54], establishing a continuous in-plane thermal conductive network shown in Figure 1(a-iii).

The versatility of the RSMO strategy extends beyond fiber-based systems to include composite films containing spherical particles and those fabricated via screen printing. As shown in Figures S1, S2, and Movie S2, within the roll pressure field, ultrasonic induces localized, high-intensity interfacial interactions between the polymer-coated particles. These interactions drive particles rearrangement and dense packing^[50,55–57], effectively disrupting the surface oxide layers and polymer coating on the particle surface^[58]. Consequently, the particles undergo thermal softening and inter-particle bonding^[50], forming a dense network that serves as an efficient channel for heat carrier transport. This densification effect is substantial, which will be further elaborated on in the subsequent sections and Supplementary Note 1.

The synergistic interaction of ultrasonic and roll pressing endows the RSMO with broad universality, enabling the mass preparation of large-area, thermally conductive composite films across diverse material systems, such as boron nitride (BN), Cu, and BN-Ga in PVA electrospinning systems (Figures 1(d-i) and S3), as well as BiInSn, GR, and Cu in paper-based screen printing (Figure S4). Specifically, RSMO achieves excellent densification and orientation at a processing speed of $50 \text{ mm}\cdot\text{s}^{-1}$ for composite films with a width of 200 mm. By optimizing the dimensions of the ultrasonic-rolling equipment, the processing width can be scaled to 500 mm or more. Additionally, this method preserves material flexibility (Figure 1(d-ii)) and facilitates the synergistic coupling of orientation and heterogeneous integration of multiple materials. Different thermally conductive fillers can achieve regionalized filler compounding and orientation simultaneously through the RSMO, which in turn modulates the heat transfer pathway, as shown in Figures 1(d-iii), (d-iv), and S5. In Figure 1(e), we benchmark the RSMO against mainstream alignment techniques^[27,46,48,49,59]. Notably, without relying on external high temperature or high pressure, RSMO can efficiently produce composites with superior thermal conductivity and mechanical strength within a timescale of seconds.

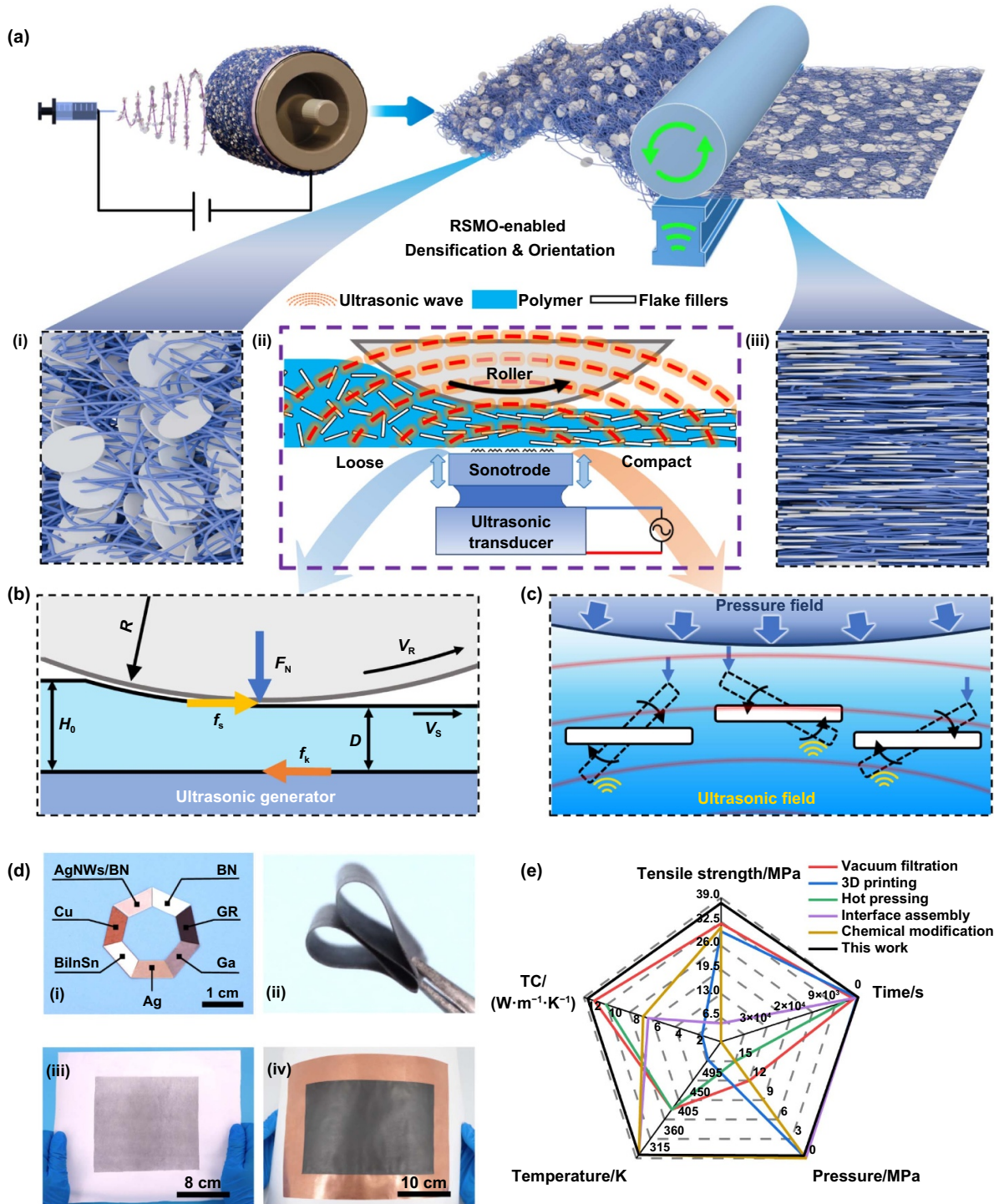


Figure 1. Descriptions of the RSMO method. (a) Oriented and compact thermally conductive films can be stably prepared via RSMO; schematic cross-sectional morphologies (i) before and (iii) after treatment, and (ii) schematic of the RSMO working principle. (b) macroscopic force analysis of samples during the RSMO process. F_N : preload, f_s : dynamic friction force, f_k : static friction force. (c) microscopic force analysis of samples during the RSMO process under optimal operating conditions. (d) digital photographs of RSMO-treated thermally conductive films. (i) photographs of samples from various material systems assembled into regular heptagons after cutting, (ii) flexibility of the treated electrospun film, (iii) large-sized PVA/BN and PVA/Ga electrospun films prepared by RSMO, and (iv) large-area paper-based Cu&GR composite thermally conductive films prepared by RSMO. (e) comparison of the tensile strength and thermal conductivity of the RSMO-processed samples versus other orientation methods, alongside the corresponding processing parameters, including temperature, pressure magnitude, and duration. (Vacuum filtration^[27]; 3D printing^[46]; hot pressing^[47]; interface assembly^[48]; chemical modification^[49]).

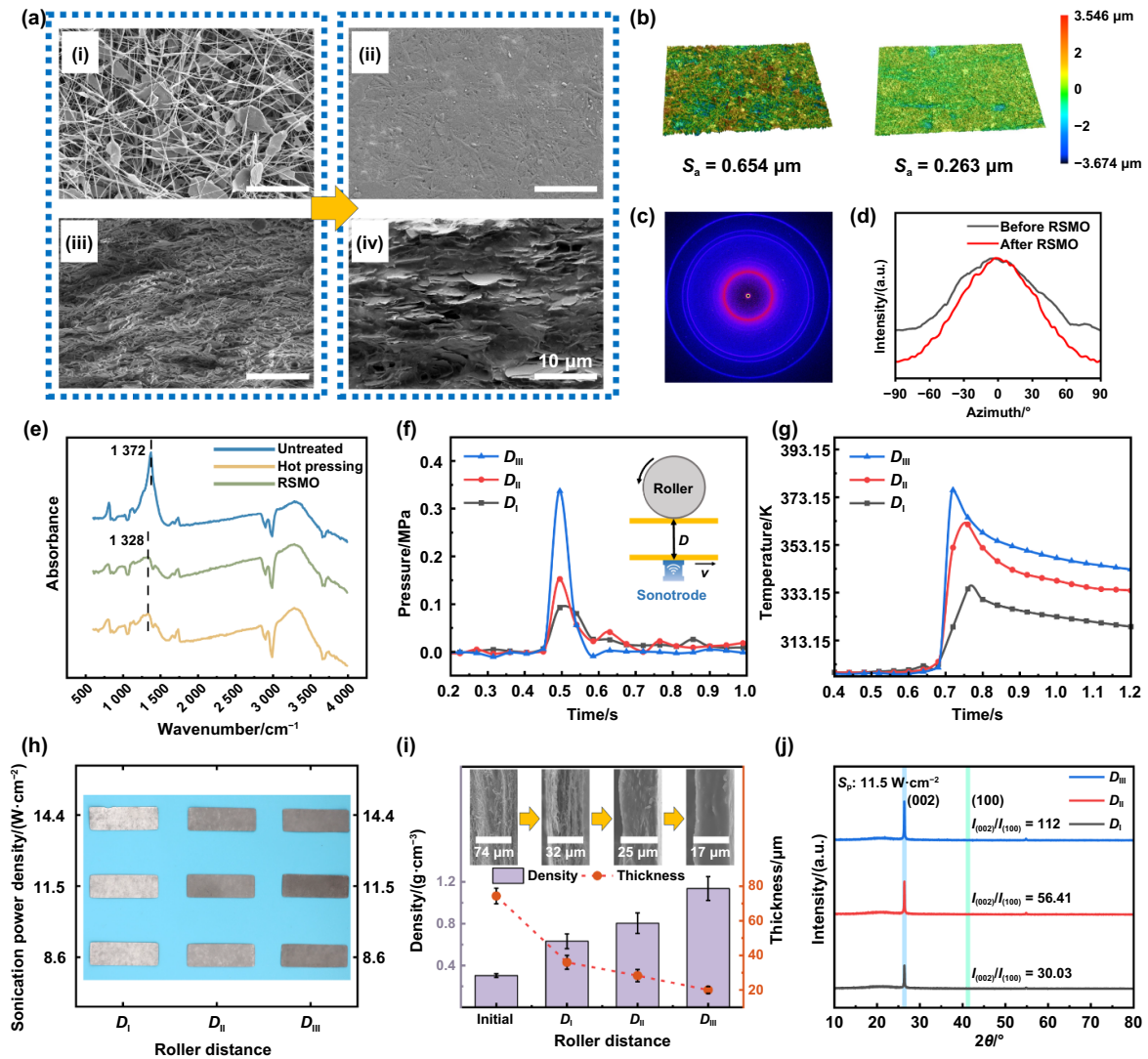


Figure 2. Characterization of dense and oriented structures (using PVA/BN-Ga composites as the model samples). (a) SEM images of the PVA/BN-Ga sample: (i) surface before RSMO; (ii) surface after RSMO; (iii) cross-section before RSMO; (iv) cross-section after RSMO. (b) Laser microscopy 3D images of PVA/BN-Ga composites before and after RSMO treatment. (c) WAXS of the PVA/BN-Ga composite. (d) The azimuth angle intensity curves of the BN characteristic crystal planes of the samples before and after RSMO treatment; (e) FTIR-ATR spectra of untreated, hot pressed, and RSMO-treated PVA/BN-Ga composites. (f) Pressure–time curves of samples processed by RSMO at different roller distances with the sonication power density fixed at $11.5 \text{ W}\cdot\text{cm}^{-2}$. The inset shows a schematic of the roller distance with a protective film. (g) Temperature–time curves of samples treated by RSMO at different roller distances with the sonication power density fixed at $11.5 \text{ W}\cdot\text{cm}^{-2}$. (h) Photographs of samples processed at different sonication power densities and roller distances. (i) Thickness and density of samples treated at different roller distances under a sonication power density of $11.5 \text{ W}\cdot\text{cm}^{-2}$. Insets show cross-sectional SEM images of the corresponding samples. (j) XRD results of samples after treatment at a fixed sonication power density of $11.5 \text{ W}\cdot\text{cm}^{-2}$ and different roller distances.

2.2. RSMO-enabled densification and orientation mechanisms

The densification and orientation mechanisms governing RSMO performance were systematically investigated using electrospun films embedded with high-thermal-conductivity nano/micro-particles (PVA/BN-Ga) as a model system. Initially, microstructural characterization (Figure 2(a)) revealed that the RSMO treatment effectively eliminated the original surface voids of the loose, porous spinning network, transforming it into a dense, solid entity. This densification

is evidenced by the significant reduction in film thickness and the increase in density (Figures S6, S16 and S17). Concurrently, the surface quality improved drastically, with the 3D surface arithmetic mean deviation (S_a) decreasing from $0.654 \mu\text{m}$ to $0.263 \mu\text{m}$ (Figure 2(b)). Internally, the random flake fillers (BNNs) were flattened and compacted in-plane, establishing a horizontally oriented structure. This orientation evolution was further corroborated by 2D wide-angle X-ray scattering (WAXS). In the 2D color map (Figure 2(c)), the diffraction signal at approximately 26.5° , corresponding to the (002) crystal plane of h-BN, exhibits a ring shape

with intensity inhomogeneity. For the samples treated by RSMO, the intensity–azimuth angle curve of this crystal plane exhibits a prominent peak at 0° and an extremely low signal at $\pm 90^\circ$, demonstrating the distinct in-plane alignment of BNNSs (Figure 2(d))^[51,60]. Additionally, Fourier transform infrared (FTIR) analysis (Figure 2(e)) revealed a red-shift and intensity decrease of the PVA methylene ($-\text{CH}_2-$) peak at $1\,372\text{ cm}^{-1}$, suggesting that the RSMO process induces molecular chain rearrangement and increased crystallinity^[61]. Interfacial hydrogen bonding between the hydroxyl groups ($-\text{OH}$) and BN may also be strengthened. Crucially, this RSMO-induced structural ordering is a universal phenomenon, consistently observed across various PVA-based thermally conductive fiber films (with fillers such as Cu, Ga, Al_2O_3 , etc., Figures S3 and S8–S11) and screen-printed films (Figures S4, S6, and S7).

To elucidate the physical drivers behind this transformation, we monitored key process parameters in real-time. Figure 2(f) illustrates that the pressure exerted on the film during the RSMO process is directly controlled by the roller distance (D). Specifically, as the roller distance decreases from D_I ($360\text{ }\mu\text{m}$) to D_{III} ($300\text{ }\mu\text{m}$) at a constant ultrasonic energy of $11.5\text{ W}\cdot\text{cm}^{-2}$, the pressure increases from 0.09 MPa to 0.33 MPa (the definition and adjustment method of the roller distance are shown in section 4.4). Simultaneously, ultrasonic vibration generates significant frictional heat (Figures 2(g) and S12). Under optimized conditions ($11.5\text{ W}\cdot\text{cm}^{-2}$, D_{III}), the film reaches a transient temperature of $\sim 376.15\text{ K}$ (temperatures under other conditions are shown in Figure S12), which is significantly higher than the glass transition temperature (T_g) of the composite (Figure S13). This temperature rise is critical because it induces thermal softening of the PVA matrix, enabling plastic deformation and facilitating the rearrangement of fillers. Notably, a high degree of temperature uniformity ($<6\text{ K}$ variance) was achieved across the lateral width of the composite during large-area processing (Figure S14). The moisture evaporation of the PVA matrix potentially induced thereby was also quantified, as shown in Figure S15.

Based on these observations, we propose a synergistic densification and orientation mechanism. As schematically illustrated in the top-right inset of Figure 1(a) (where the primary working region of RSMO acting on the film can be defined as a linear contact zone), RSMO creates a unique line-contact loading zone where pressure and thermal energy are highly localized (as illustrated by the schematic representations of the ultrasonic action zone in Figures 1(a)–(c), the labeled compressive force F_N in Figure 1(b), and the delineated downward pressure field region in Figure 1(c)). High-frequency ultrasound induces intense interfacial friction, rapidly heating the matrix above T_g . The heated matrix exhibits a certain degree of fluidity, and the internal particles undergo thermal softening and relative slippage^[55,56]. Under the synergistic effect of vertical ultrasonic vibration and mechanical pressure, both the polymer and particles undergo interfacial compatibility and plastic deformation, which effectively compresses and fills the internal voids to achieve densely packed configuration^[50] (as revealed by the cross-sectional morphology schematic of films

after RSMO treatment, intertwined PVA fibers and randomly distributed BN particles (Figure 1(a-i)) evolve into highly aligned PVA fibers and oriented BN platelets (Figure 1(a-iii))). Under the interaction of the roll-pressing-ultrasonic coupled field, the randomly distributed fillers inside the composite rotate as a result of the superposition of vertical downward pressure and periodic acoustic stress. When a balance between the two forces is achieved, the fillers achieve favorable horizontal alignment (more detailed mechanism analysis is provided in Supplementary Note 1).

Thermoplastic PVA serves as the core material in the RSMO process, and its performance varies significantly with morphology and concentration. In fiber-rich electrospun films, PVA serves a dual purpose: it acts as a structural scaffold and promotes ultrasonic-induced softening and rheological flow. This behavior facilitates the redistribution of thermally conductive fillers^[62,63] and decreases the torsional orientation torque^[54], thereby streamlining the densification process^[52,53]. Furthermore, its thermally softened bonding characteristics ensure the stability of particles during the RSMO process. In contrast, in screen-printed systems where PVA exists as a thin coating, it primarily ensures the pre-fixation of particles (as shown in Figures S1 and S2). Under ultrasonic excitation, this interfacial layer acts as a viscoelastic buffer that absorbs kinetic energy, dampening rigid inter-particle collisions and optimizing the stress distribution on particle surfaces to achieve stable densification^[58] (Figure S4, metallic luster in practical imaging confirms this favorable microstructural evolution). Without this protective layer, the fillers suffer from unbuffered mechanical shocks that hinder densification, leading to poor consolidation (detailed comparative analysis is available in Figure S7 and Supplementary Note 1).

The structural quality is highly sensitive to the processing parameters. As presented in Figures 2(h) and S11, samples subjected to higher preload and sonication power density (S_p) exhibited superior densification, characterized by a darker, more uniform surface color and smoother morphology. This is attributed to the tight interconnection of fibers and the homogeneous distribution of liquid metal (the specific definition of sonication power density is provided in section 4.4). However, at higher sonication power densities, the intense ultrasonic thermal effect leads to particle aggregation and exudation, thereby possibly disrupting the uniformity of the thermal network.

Under optimized conditions, the RSMO process reduced the thickness of the electrospun composites from $74\text{ }\mu\text{m}$ to $17\text{ }\mu\text{m}$ and increased their density from 0.31 to $1.23\text{ g}\cdot\text{cm}^{-3}$ (Figure 2(i)). Similar densification results were consistently observed in other material systems and film-forming processes (Figures S16 and S17), demonstrating the effectiveness of roller distance as a key regulatory parameter.

To further evaluate the influence of the processing parameters on the composite's microstructure, XRD was employed to quantify the BNNS orientation. The alignment efficiency is reflected by the peak intensity ratio of the (002) plane to (100) plane (I_{002}/I_{100})^[64]. As indicated in Figure 2(j), reducing the

final roller distance from D_I to D_{III} significantly enhances the parallel orientation of BNNSs, evidenced by the increase in the I_{002}/I_{100} ratio from 30.03 to 112. Conversely, excessive energy density results in a diminished orientation (Figures 2(j) and S18), attributed to intense vibrations causing over-twisting of the BNNSs and disrupting their orderly arrangement.

To recap the key effects of sonication power density, the temperature rise is insignificant, and the superposition of acoustic and compressive stresses is insufficient; consequently, reducing the roller distance primarily induces vertical displacement of the fillers, hindering effective rotational alignment. At a moderate power density ($11.5 \text{ W}\cdot\text{cm}^{-2}$), the ultrasonic thermal effect appropriately softens the polymer matrix, whereas stress superposition amplifies filler compression and torsion; combining this with the minimum roller distance (D_{III}) yields optimal alignment and densification. However, at excessively high power density ($14.4 \text{ W}\cdot\text{cm}^{-2}$), intense heat triggers the melting and exudation of the liquid metal. Furthermore, the stress superposition becomes imbalanced as ultrasonic forces overwhelm mechanical pressure, causing excessive filler torsion and significantly deteriorating the alignment performance (Supplementary Note 1).

In conclusion, the RSMO process with optimized parameters (D : D_{III} , S_p : $11.5 \text{ W}\cdot\text{cm}^{-2}$) achieves controllable densification and orientation by coupling ultrasound with roller pressing, leveraging the vibration-induced frictional heating effect and the synergistic interaction between ultrasound and mechanical pressure to induce softening, deformation, and oriented alignment of the composite materials.

2.3. RSMO-enabled highly thermal conductivity

Based on the densification and orientation mechanism of the RSMO, fillers within the polymer matrix are engineered to form a highly efficient thermal conductive network, enabling this process to remarkably enhance the thermal conductivity of composites. This enhancement is universally applicable to various filler types, including two-dimensional flakes, spherical particles, one-dimensional materials, and multi-component systems (Figure 3(a), with filler contents detailed in Figure S19 and Table S1). The thermal conductivity of each material system is governed by two factors: the inherent conductivity of the fillers and the structure of the thermally conductive network formed during RSMO treatment. For two-dimensional fillers (e.g., BNNSs), compared with pure PVA, RSMO promotes the formation of overlapping and interconnected in-plane continuous channels, which increases the in-plane thermal conductivity of PVA/BN composites ($11.54 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) by 42 times. For spherical fillers (e.g., Al_2O_3), the dense arrangement of Al_2O_3 particles facilitates the formation of partially interconnected heat-conductive pathways, resulting in a thermal conductivity of $5.07 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ for the PVA/ Al_2O_3 composites after RSMO treatment (Figure S8). Notably, despite its lower intrinsic conductivity than Al_2O_3 , room-temperature liquid metal gallium exhibits excellent deformability, allowing it to conform and fuse into continuous conduction paths, thereby achieving a superior conductivity of $7.53 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [65,66]. Beyond single filler

systems, RSMO demonstrates exceptional efficacy in regulating thermal conductivity in hybrid filler systems. In the PVA/BN-Ga composite, Ga undergoes a melting transition under the RSMO thermal field and infiltrates the micro-gaps between the BNNSs and the PVA matrix via cyclic extrusion. This liquid metal bridging effect interconnects adjacent BNNSs both horizontally and vertically, creating a robust “brick and mortar” interfacial structure (Figure S9). The multi-dimensional interface modulation mechanism effectively optimizes heat conduction paths at low filler loadings, significantly enhancing both in-plane ($12.67 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) and out-of-plane thermal conductivities [35] ($0.61 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$). Conversely, adding BiInSn has the opposite effect, likely due to its high melting point, poor deformability, and low thermal conductivity, which undermine the construction of BN heat transfer networks. Furthermore, to verify the method’s versatility, 1D silver nanowires (AgNWs) were introduced to bridge other fillers. Driven by RSMO-induced vibration and extrusion, the AgNWs formed intimate contacts with spherical or flake-shaped fillers (Figure S10). The PVA/ Al_2O_3 -AgNWs and PVA/BN-AgNWs composites achieve in-plane conductivities of 7.94 and $12.15 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, respectively.

In addition, RSMO-treated composites exhibited significant anisotropy, with anisotropy indices exceeding 20, as shown in Figure 3(b). During RSMO, the thermally conductive filler undergoes plastic flow and cyclic stress, leading to in-plane slippage and alignment. This phenomenon results in discontinuous stacking of the material in the vertical direction and creates a fragmented heat conduction path, thereby exacerbating the disparity in thermal conductivity between the in-plane and out-of-plane directions. The thermal stability of the composites was verified over 10 heating–cooling cycles, showing negligible degradation even at extreme temperatures (Figure S20). For the processing of large-area composites ($200 \text{ mm} \times 250 \text{ mm}$, Figure 3(c)), RSMO enables highly uniform transverse thermal conductivity and thickness, with the overall coefficient of variation (CV) maintained below 5%.

To further evaluate the thermal conductivity of the RSMO-treated composites, we investigated the effects of different processing parameters. As shown in Figure 3(d), the peak thermal conductivity was attained at a final roll gap of D_{III} and an ultrasonic power density of $11.5 \text{ W}\cdot\text{cm}^{-2}$, simultaneously maximizing density and orientation. Furthermore, the cushion film showed negligible influence on conductivity (Figure S21). Next, a finite element simulation was conducted to visually compare the heat-transfer behavior in the composites (Figure 3(e); additional details are provided in Supplementary Note 2). Compared with the initial random distribution, the RSMO-treated fillers formed well-aligned, horizontally interconnected conduction chains, significantly improving the heat transfer efficiency from the heat source to the cold side.

The introduction of ultrasound is a critical mechanism for enhancing thermal conductivity in the RSMO process. Unlike simple low-pressure rolling, which leaves interfacial voids that introduce significantly high thermal resistance, ultrasound provides particles with sufficient energy to overcome

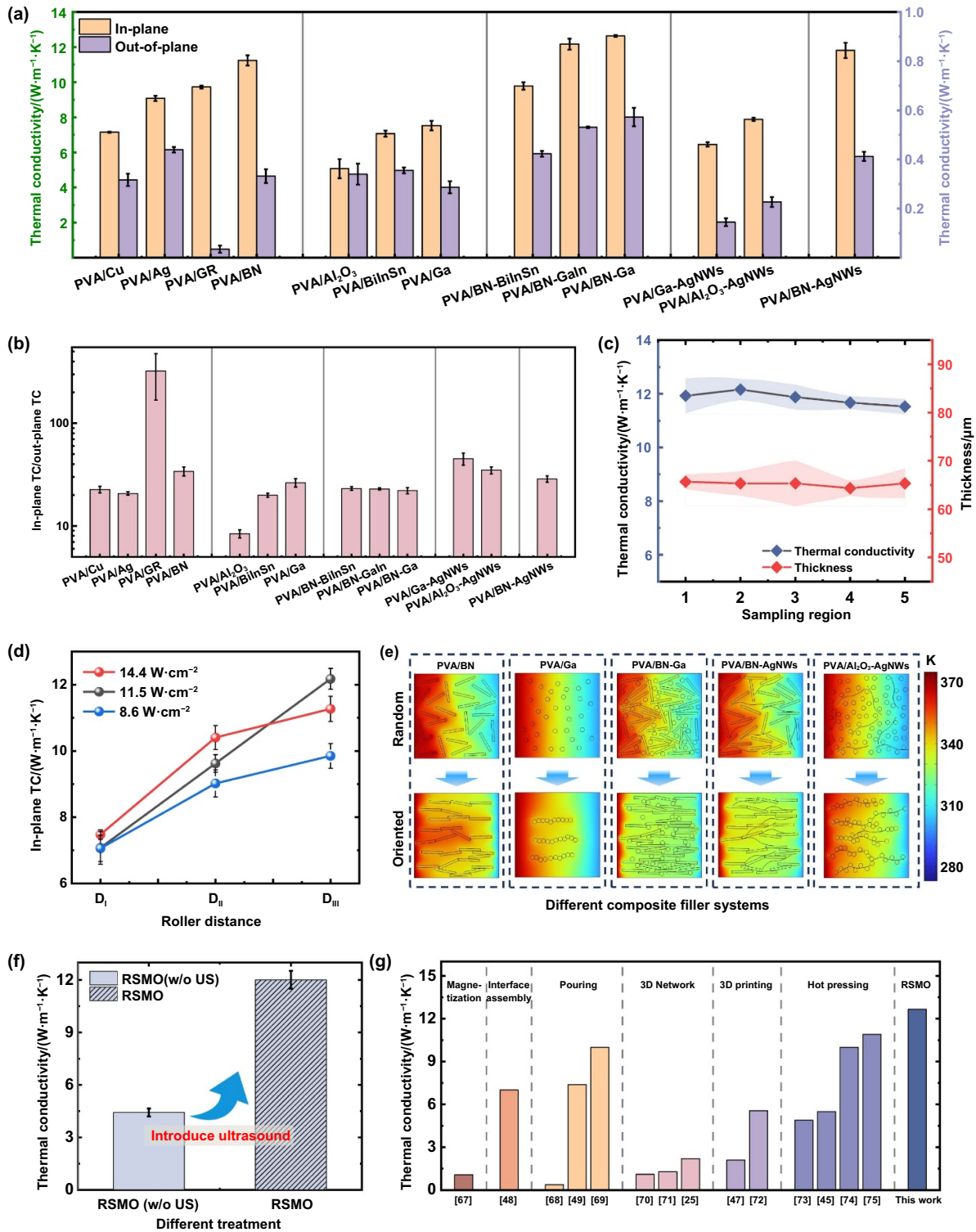


Figure 3. Characterization of thermal properties. (a) In-plane and out-of-plane thermal conductivity of multiple thermally conductive material systems (including flake, wire, sphere, and their hybrid systems) after RSMO treatment. (b) Anisotropic thermal conductivity of nanocomposites after RSMO treatment. (c) Uniformity of thermal conductivity and thickness across a large-area composite film after RSMO treatment. (d) Effect of different sonication power densities and roller distances on the in-plane thermal conductivity of samples. (e) Finite element simulation results comparing the thermal conductivity of various composites in randomly distributed versus oriented states. (f) Significant enhancement of thermal conductivity achieved by introducing ultrasonication compared with simple roller pressing (0.4 MPa, 363.15 K, 50 $\text{mm}\cdot\text{s}^{-1}$). (g) Comparison of the thermal conductivity of BN composites prepared by different methods reported in the literature.

inherent surface forces, promoting dense packing and orientation. Specifically, interfacial frictional heating softens the polymer matrix and induces particle slip and rearrangement. Periodic ultrasonic impacts, acting in tandem with mechanical pressure, synergistically facilitate the structural densification of the composite film. Concurrently, the coupling of acoustic and mechanical stresses induces a localized torque that triggers the reorientation of internal fillers, promoting highly aligned horizontal architectures. Consequently, a continuous and compact in-plane thermally conductive network is established, providing highly efficient pathways for phonon transport (Supplementary Note 2).

Under low working pressure, ultrasonic treatment boosts the thermal conductivity by approximately 171% (Figure 3(f)), achieving an amplification effect comparable to that of long-term high-temperature and high-pressure hot pressing^[45] (Figure S22). Preliminary calculations estimate of RSMO's production capacity is $36 \text{ m}^2 \cdot \text{h}^{-1}$, which is approximately two orders of magnitude greater than that of conventional hot pressing, demonstrating the great potential of RSMO for rapid, large-scale production.

Finally, Figure 3(g) and Table S2 systematically benchmark the thermal conductivity of composites with similar BN contents prepared via various processes reported in recent literature, including magnetic field orientation^[67], interface assembly^[48], pouring^[49,68,69], 3D network^[25,70,71], 3D printing^[47,72], and hot pressing^[45,73–76]. The in-plane thermal conductivity of our PVA/BN-Ga composite film ($12.67 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$) surpasses that of the composites prepared by all the previously mentioned approaches. These results highlight that RSMO stands out as a novel, efficient, and scalable approach for enhancing the thermal conductivity of polymer-based thermal conductive composites.

2.4. RSMO-enabled high flexibility

In addition to thermal conductivity, the mechanical robustness of composites is paramount for practical applications. Figure 4(a) highlights the outstanding flexibility, deformability, and tensile strength of the RSMO-processed PVA nanocomposites. The films can be effortlessly folded into complex shapes, such as a windmill or an origami crane (Figure 4(a-i) and (a-iii)). Notably, even after 180° folding, the interface maintains a dense connection without fracture, as evidenced by the cross-sectional SEM image (Figure 4(a-iv)), illustrating sufficient flexibility. Furthermore, RSMO endows the composites with excellent load-bearing capacity: the sample ($10 \text{ mm} \times 30 \text{ mm} \times 0.06 \text{ mm}$) can withstand a static load of 1 kg without failure (Figure 4(a-ii)).

As shown in the stress–strain curves (Figure 4(b)), all RSMO-treated composites exhibit tensile strengths exceeding 20 MPa, demonstrating a superior ability to adapt to extreme working conditions. In contrast, the fibers within the electrospun membrane treated under low pressure randomly entangle with weak inter-fiber bonding, resulting in a maximum tensile strength of no more than 7 MPa (Figure S23). This diversity arises because RSMO effectively facilitates the melting and interfacial adhesion of PVA fibers,

transforming the material from a fluffy fibrous state to a dense solid structure while enhancing the interfacial bonding strength of the matrix-filler (Figure 2(e)). The experimental results and analysis indicate that composite densification considerably increases the mechanical strength and structural stability of electrospun nanocomposites.

To elucidate the fracture mechanism of the RSMO sample, the surface and cross-sectional morphology of the stretched film are shown in Figures 4(b-i) and (b-ii), respectively. The poststretching surface morphology reveals increased porosity at the BN–PVA interfaces, indicating a tendency for crack initiation, whereas the interfaces in other areas remain intact. A cross-sectional analysis shows that BNNSs are uniformly layered, with the PVA exhibiting fracturing. To this end, a simplified tensile fracture model is proposed (Figures S24 (a) and (b)). Despite the RSMO treatment, the films retain residual gaps that act as initial sites for the formation of microcracks. As the tensile stress intensifies, the closely arranged PVA fibers gradually overcome the interfacial bonding force with the BNNS surface, accompanied by significant sliding and plastic deformation, ultimately exceeding the tensile limit and fracturing. These microdefects subsequently propagate, manifesting macroscopically as cracks that elongate and extend until the ultimate failure of the film occurs.

Given that these residual defects act as the principal catalysts for tensile failure, minimizing them through process optimization becomes critical for enhancing mechanical performance. Experimental evidence corroborates this: under optimal densification and orientation conditions (sonication power density: $11.5 \text{ W} \cdot \text{cm}^{-2}$, roller distance D_{III}), the PVA/BN-Ga composite reaches a peak tensile strength of 36.25 MPa (Figure 4(c)). This confirms a definitive causal link: microscopic densification and orientation directly determine macroscopic thermal and mechanical performance.

Beyond tensile strength, the long-term service performance of thermal conductive composites, such as durability and fatigue resistance, is highly important. As shown in Figure 4(d), we investigated the thermal conductivity stability of the PVA composite film. After 5 000 bending cycles (bending radius $< 1 \text{ mm}$), the attenuation rate is $\sim 9.6\%$. Furthermore, additional environmental stability tests revealed remarkable resilience (Figure S25). After thermal aging (368.15 K for 5 days), the thermal conductivity increased by approximately 10%, attributed to a secondary annealing effect that released internal stress and removed moisture. Even after prolonged exposure to high humidity (98% RH), the conductivity decreased by only $\sim 4.6\%$, proving that the densified network effectively resists moisture ingress.

To further validate the advantages of RSMO, we compared the flexural behaviors and tensile fracture properties of the composites prepared by hot pressing (15 MPa, 373.15 K, 30 min) with those fabricated via RSMO. As illustrated in Figure 4(e), the hot-pressed PVA/BN (50 wt%) composites undergo brittle fracture upon folding, whereas the structural integrity of the RSMO-treated samples is maintained. A simplified bending fracture model (Figures S24(c) and (d)) explains this divergence. Hot pressing involves prolonged exposure to high temperatures and high

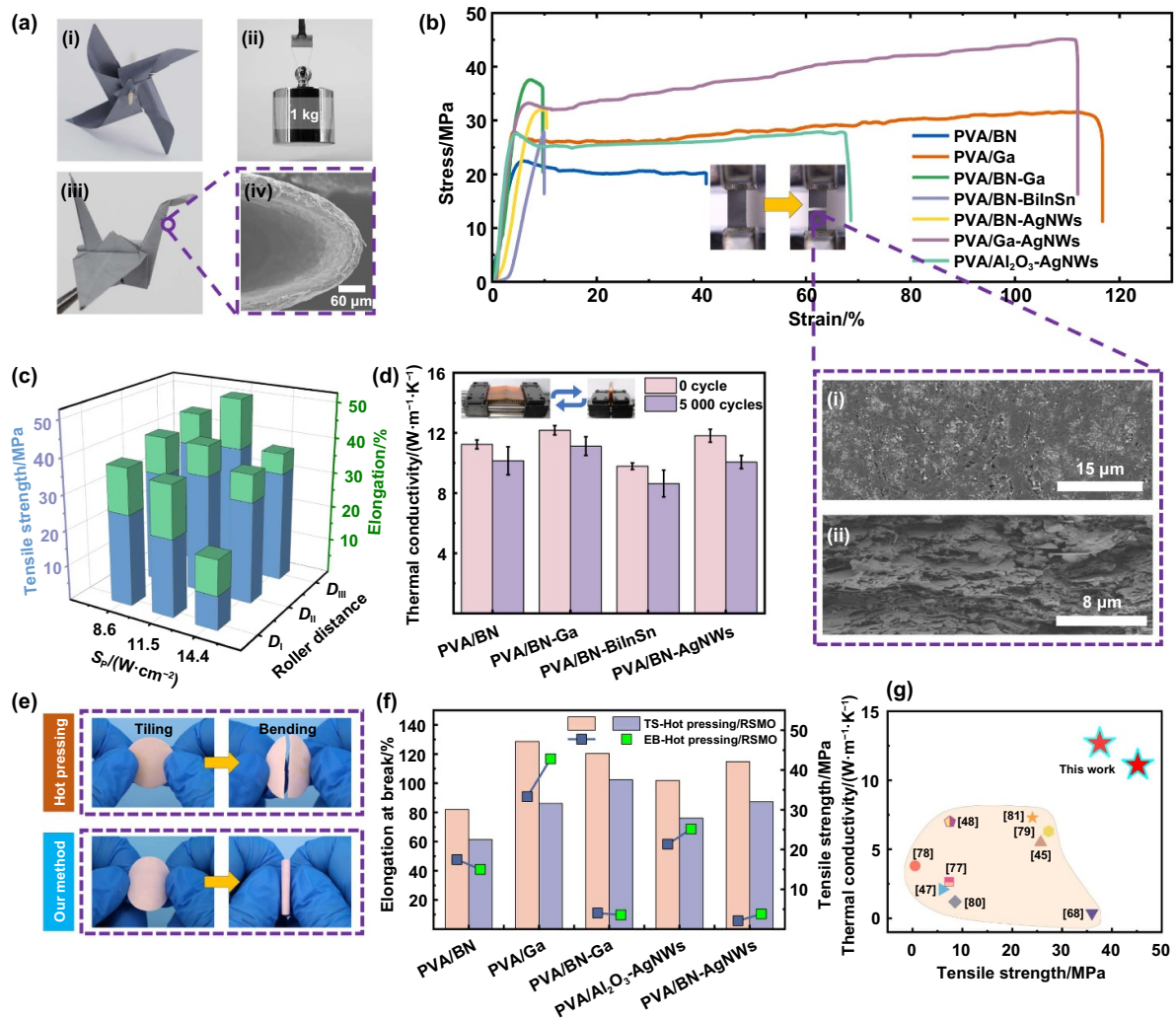


Figure 4. Characterization of flexibility. (a) RSMO-prepared PVA/BN-Ga thermal conductive films show excellent (i) flexibility, (ii) tensile properties, and (iii) foldability; (iv) SEM image of the 180° folded area. (b) Stress–strain curves of various thermally conductive material systems after RSMO treatment, alongside SEM images of (i) surface and (ii) cross-section of fractured samples; insets show digital photographs of PVA/BN-Ga nanocomposites before and after fracture. (c) Effect of different sonication power densities and roller distances on the tensile strength and elongation at break of samples. (d) Comparison of in-plane thermal conductivity changes for multiple thermally conductive material systems before and after 5 000 bending cycles; insets show digital photographs of the sample in folded and unfolded states. (e) Morphology comparison of PVA/BN (50 wt%) samples after hot pressing (15 MPa, 373.15 K, 30 min) and RSMO treatment when bent at 180°. (f) Average tensile strength and elongation at break of nanocomposites after hot pressing and RSMO. (g) Comparison of thermal conductivity and tensile strength for low-filler-content composites fabricated using various orientation processes.

pressures, conditions that tend to cause excessive polymer crosslinking and chain scission, significantly increasing brittleness^[45]. In contrast, samples prepared by RSMO are subjected to only instantaneous loads and thermal effects, resulting in minimal internal stress concentration. That is, the presence of residual micro-nano voids in RSMO samples provides a stress-relieving mechanism that enhances flexibility.

When evaluating the tensile properties, the nanocomposite prepared via high-temperature hot-pressing, as shown in Figure 4(f), demonstrates a significant improvement in its in-plane load-bearing capacity (average tensile strength: 30 MPa) due to its ultra-densification and extensive three-dimensional crosslinked network. Conversely, RSMO-treated samples are

processed under milder conditions of lower temperature and pressure, which inevitably induce minor structural defects. While these drawbacks may increase the fracture risk under tensile stress (with an average tensile strength of 24 MPa), they also impart superior resistance to bending deformation (Figure S24). Hence, RSMO represents a strategic compromise in material property regulation: moderately diminishing densification and tensile strength in exchange for preserving the requisite flexibility and foldability. Despite this, as we can see from Figure 4(g) and Table S3, the tensile strength and thermal conductivity of RSMO-prepared thermally conductive composites remain predominantly superior to those of analogous materials prepared by other orientation methods^[25,45,47–49,68,77–81].

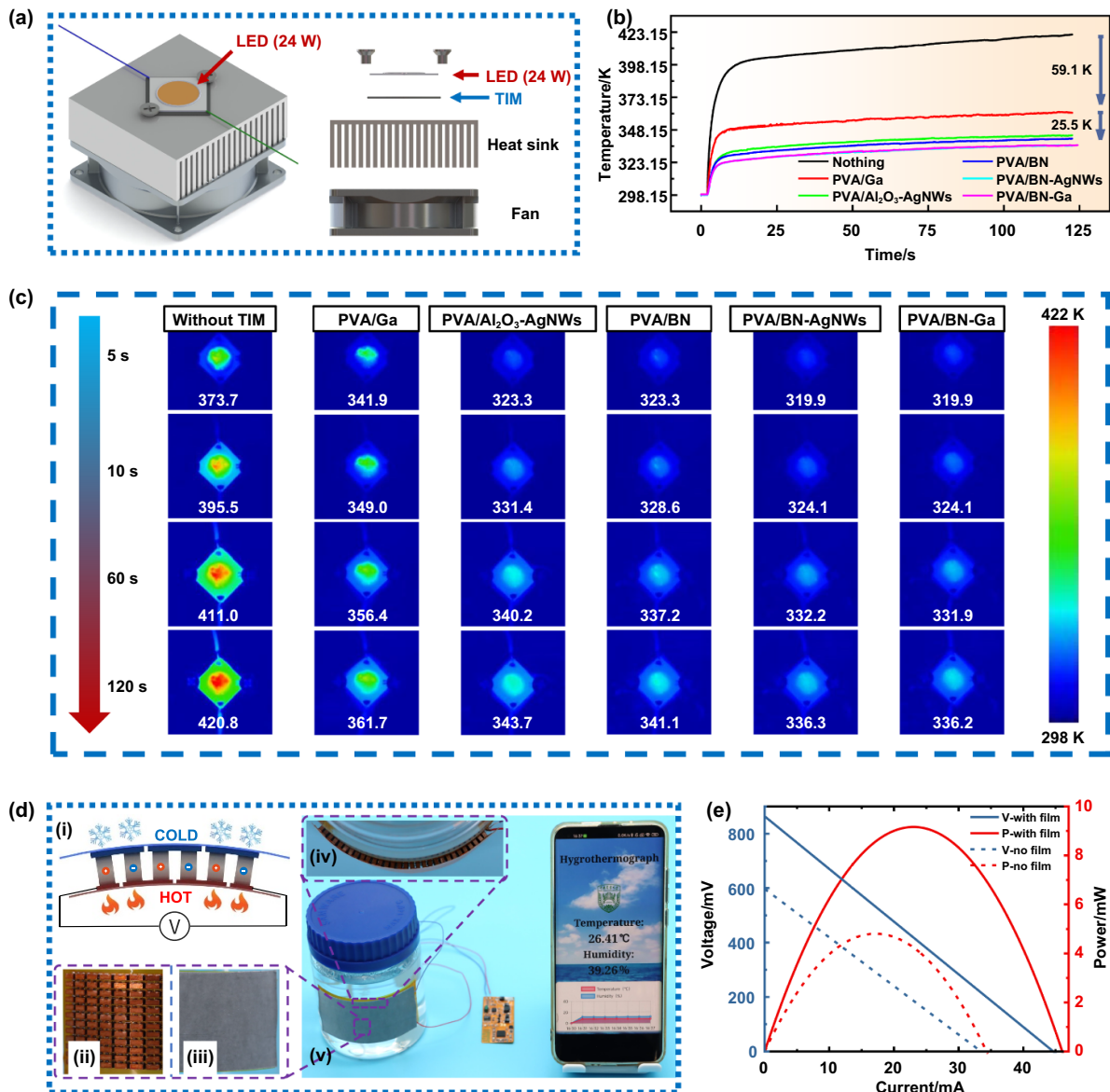


Figure 5. Application of composite films in LEDs and thermoelectric devices. (a) Schematic of the high-power LED cooling system. (b) Surface temperature curves of LEDs with different thermally conductive composites as TIMs. (c) Thermal infrared images of the LED surface with different thermally conductive composites as TIMs. (d) Installation of thermally conductive films for thermoelectric devices. (i) Working principle of the thermoelectric device; (ii) photograph of the thermoelectric device without and (iii) with an RSMO-treated film; (iv) top view of RSMO-treated PVA/BN-Ga films attached to the cold end of the thermoelectric device; (v) electricity generated by the thermoelectric device powers a temperature and humidity sensing circuit, with sensor data transmitted to a mobile device via Bluetooth. (e) $V-P-I$ curves of thermoelectric devices with and without a thermally conductive film attached to the cold end.

2.5. Applications

Owing to the unique combination of high thermal conductivity, robust mechanical strength, and superior flexibility, RSMO-processed composites are promising candidates for next-generation flexible thermal management. To validate their practical efficacy, we conducted proof-of-concept demonstrations in two critical scenarios: high-power electronic cooling and wearable energy harvesting.

Firstly, the composite was employed as a thermal interface material (TIM) between a high-power LED chip (24 W) and a heat sink, with the surface temperature monitored

by infrared thermography (Figure 5(a)) (all the TIMs were tested by a multimeter before use and were found to be non-conductive). In the absence of a TIM, the LED temperature increased dramatically within 5 seconds, reaching 420.75 K after 120 seconds due to interfacial air resistance (Figures 5(b) and (c)). Remarkably, the integration of the RSMO-prepared films significantly mitigated this overheating. The balanced temperature decreased to 361.65 K for PVA/Ga and 336.15 K for PVA/BN-Ga, with respective temperature reductions of 59.1 K and 84.6 K. Consistent with their intrinsic thermal conductivities, other RSMO composites also demonstrated superior heat dissipation, confirming their

capability to meet the rigorous thermal demands of high-power electronics.

To demonstrate its applicability in wearable electronics, the PVA/BN-Ga film was integrated onto the cold side of a flexible TEG (35 mm × 70 mm × 3.2 mm) that operated based on the Seebeck effect (Figure 5(d-i))^[82], with a beaker containing boiling water as the hot source (Figure 5(d-v)), P/N-type thermoelectrical legs integration follows previous methods^[58,83]. The TEG harvested thermal energy to power a wireless temperature–humidity sensing system. Under natural convection (ambient temperature: 298.15 K), the efficient heat dissipation provided by the RSMO film maintained a large temperature gradient across the TEG, ensuring stable real-time data transmission to a mobile app via Bluetooth (Figure 5(d-v)).

Quantitative evaluation (Figure 5(e)) revealed that the device equipped with the RSMO heat-dissipation film generated an open-circuit voltage (V_{open}) of 863 mV and an output power (P_{out}) of 9.16 mW. These values represent significant enhancements of 1.44-fold and 1.9-fold, respectively, compared to the bare device. In the constant resistance test mode (Figure S26), the closed-circuit voltage of the thermoelectric device (V_{in}) reached 0.31 V with the film, representing a 15% enhancement over the film-free case (0.27 V). The successful implementation of RSMO-prepared thermally conductive films in LEDs and thermoelectric devices not only verifies their excellent heat transfer, heat dissipation performance, and flexibility but also unveils the vast application potential of RSMO in the realm of flexible electronic thermal management.

While RSMO presents distinct technical merits for continuous manufacturing, it still faces several foreseeable challenges in scaling up from laboratory-scale to industrial-scale operations. These hurdles include acoustic pollution generated during operation, overheating of ultrasonic transducers induced by prolonged working periods, and the design of ultrasonic transducers capable of delivering uniform ultrasonic energy over larger processing areas. Nevertheless, in contrast to vacuum filtration (which is plagued by low efficiency and size constraints) or magnetic orientation (which requires bulky external magnetic fields), RSMO integrates shear force and ultrasound into a continuous winding system, thereby offering a more promising avenue for the roll-to-roll large-scale fabrication of high-performance thermal management materials.

3. Conclusions

In summary, the RSMO method innovatively combines high-frequency ultrasonic vibration with rotary roller compaction technology. The thermo-mechanical effect generated by the coupled ultrasonic and pressure fields induces softening and adhesion of the matrix along with the horizontal alignment of the fillers in it. This entire process is completed within one second. The resulting densified and oriented microstructure establishes a continuous in-plane thermal conductivity

pathway, diminishes the defect density, and achieves excellent anisotropic thermal conductivity, tensile strength, and flexibility. This process can be applied to a variety of thermally conductive filler systems. Its unique line contact loading method facilitates gentle orientation while reducing the processing time, rendering it compatible with large-scale production. As a demonstration, thermally conductive films prepared via RSMO exhibit superior flexibility and thermal management properties in high-power LED modules and flexible thermoelectric generators (TEGs). This novel orientation-controlled method provides a new technical path for the preparation of thermal management materials and holds relatively broad industrial application prospects.

4. Methods

4.1. Materials

All the materials and chemicals used in this work were commercially available. BNNSs with an average diameter of 3 μm were obtained from Suzhou Nutpool Materials Technology Co., Ltd. Bi (ingots, 99.99%), Ga (ingots, 99.99%), In (ingots, 99.995%), and Sn (beads, 99.99%) were employed to prepare eutectic and Ga-based LMs. They were combined at predefined weight ratios and melted at 473.15 K for 10 h to obtain eutectic BiInSn (Bi: In: Sn = 31.6: 48.8: 19.6), GaInSn (Ga: In: Sn = 68.5: 21.5: 10), and GaIn (Ga: In = 75.5: 24.5) alloys. Cu flakes (10 μm in diameter), Ag flakes (3 μm in diameter), and Al₂O₃ particles (1 μm in diameter) were purchased from Shanghai Maoguo Nano Technology Co., Ltd. Graphene dispersion (PVP dispersion, 1 mg·mL⁻¹, ethanol solvent) were provided by Suzhou Tanfeng Graphene Technology Co., Ltd. Silver nanowires (20 mg·mL⁻¹, ethanol solvent) applied in this work were purchased from Jiangsu Xianfeng Nanomaterials Technology Co., Ltd. PVP (polyvinyl pyrrolidone, Mn = 1 300 000), PVA (217, Kuraray, Japan), and ethanol (99.7%) were purchased from Macklin Biochemical Co., Ltd. Flexible substrates of A4 paper (160 g) and PI (polyimide) with a thickness of 125 μm were purchased from Taobao Mall.

4.2. Preparation of nano/microparticle inks

A mixture of 1 g PVP and 99 g ethanol was heated to 358.15 K and stirred for 48 h to prepare a 1% PVP-ethanol solution. BiInSn was melted at 343.15 K and combined with a 1 wt% PVP-ethanol solution (2 g of alloy: 1 g of solution), after which the mixture was heated to 343.15 K. Subsequently, a cell ultrasonic disruptor (SFX550, Branson) was operated at an amplitude of 40% (2 s on/2 s off) to ultrasonicate the mixture for 4 minutes; then, the test tube was removed, stirred uniformly, and allowed to cool moderately. This ultrasonic operation was repeated 10 times. Similarly, 1 g of BNNSs and 5 g of 1% PVP-ethanol solution were prepared following the same procedure, yielding an ethanol dispersion of the BNNSs. EGaIn/Ga ink was prepared by sonicating 2 g of liquid metal in 1 g of 1%

PVP-ethanol solution in an ice water bath for 20 min at an amplitude of 40% (2 s on and 2 s off). Similarly, copper/silver flake inks were prepared using the same ratio and process. The digital photos are shown in Figure S27.

4.3. Preparation of thermal conductive films

For the electrospinning process, a 10% PVA solution was prepared by stirring 1 g of PVA into 9 g of deionized water at 363.15 K for 48 h. The precursor solution for electrospinning was prepared by mixing the thermally conductive material dispersion and a PVA aqueous solution at a certain mass ratio, followed by stirring at 800 rpm for 12 hours using a magnetic stirrer. After defoaming, the solution was poured into a syringe and electrospun at a propulsion rate of $1 \text{ mL} \cdot \text{h}^{-1}$, with an applied voltage of 16 kV and a collection distance of 15 cm. The collection device was a metal drum covered with silicone oil paper and rotated at 300 rpm. The electrospinning proceeded at room temperature with a surrounding humidity of 40%. Throughout the spinning process, the syringe was rotated every 2 h to prevent the settling of the nano/microparticles in the solution.

For the screen printing process, the PVP-coated particles were centrifuged at 8 000 rpm for 5 minutes, and the supernatant was removed and dried at 353.15 K for 6 hours. A predetermined amount of 5% PVA aqueous solution was then added, and the particles were sonicated for 5 minutes at a 20% amplitude (2 s on and 2 s off). The inks were printed on the surface of the A4 papers (160 g) using the screen-printing technique and subsequently dried at room temperature for 2 hours.

4.4. The orientation and densification process of thermal conductive films

The ultrasonic roller pressing device consists of a roller (diameter: 10 cm; width: 320 mm), a sonotrode (15 kHz; 1.2 kW), and a motor (Figure S28). The roller speed (v) is $50 \text{ mm} \cdot \text{s}^{-1}$. The sonotrode has a working face area of 80 cm^2 (dimensions: length 32 cm $\times$ width 2.5 cm). The ultrasonic instrument's output power (P) was constrained to approximately 600–1 200 W, with three settings investigated: 690, 920, and 1 150 W. To account for differences in sonotrode size and preparation scale, the ultrasonic power was normalized to the corresponding sonication power density (S_p) of 8.6, 11.5, and $14.4 \text{ W} \cdot \text{cm}^{-2}$, respectively. The specific calculation formula is $S_p = P/A$, where P is the output power and A is the area of the sonotrode's working face.

The roller distance (D) is defined as the interval separating the roller from the sonotrode. For a film with an initial height denoted as H_0 , the roller distance is set as the sum of the protective film thickness (H) and the target final film thickness. On the basis of previous experimental experience, three different distances were investigated: $D_I = H + 0.4H_0$, $D_{II} = H + 0.3H_0$, and $D_{III} = H + 0.2H_0$. To prepare a film with a target thickness of $60 \text{ } \mu\text{m}$ as an example, a paper sheet with a total thickness of $240 \text{ } \mu\text{m}$ was selected as the backing

film. The thickness of the original laminated electrospun film was approximately $300 \text{ } \mu\text{m}$, and the specific D_I , D_{II} , and D_{III} were 360, 330, and $300 \text{ } \mu\text{m}$, respectively.

Specific operation process of RSMO: the electrospinning film was removed from the silicone oil paper and stacked to the required thickness. The stacked spinning membrane was placed between two pieces of paper or polyimide films, and the entire sandwich structure was positioned between the drum and the sonotrode, leaving a small part of the front end exposed for subsequent traction. The distance between the roller and the sonotrode was set using a dial gauge. After the power and the roller distance of the ultrasonic system were adjusted as needed, the instrument was started, and the roller rotated while the ultrasonic system was operating. Afterward, the entire sandwich structure was pulled through the gap to achieve orientation and compactness. Moreover, analogous processing strategies were employed to treat paper-based thermal management films under identical operational conditions.

Production capacity of RSMO (RSMO-PC): $\text{RSMO-PC} = v \times W$, where v is the roller speed and W is the width of the processed film. Based on 20-cm-wide continuous films, the production capacity of the RSMO process reaches approximately $36 \text{ m}^2 \cdot \text{h}^{-1}$.

4.5. Characterization

The microstructures of the PVA nanocomposites were observed using scanning electron microscopy (SEM; Zeiss Sigma 360). The chemical composition of the PVA/BN-Ga was analyzed by Fourier transform infrared (Perkin-Elmer Spectrum GX) spectroscopy. The orientation of the BNNSs in the PVA/BNNS composites was analyzed by XRD (PERSEE, XD-3). The surface roughness of the sample was obtained using a laser microscope (KEYENCE VK-X150). X-ray diffraction patterns were obtained utilizing a wide-angle X-ray diffraction (WAXD, Rigaku, Japan) system. The thermal conductivities (λ) of the nanocomposites were calculated according to the equation $\lambda = \alpha \times C_p \times \rho$, where ρ denotes the density, C_p denotes the specific heat capacity, and α is the thermal diffusivity. Density ρ was calculated on the basis of weight and volume. C_p was measured by a differential scanning calorimeter (DSC214, NETZSCH). The value of α was obtained using a thermal conductivity meter (LFA 467, NETZSCH). The thermal conductivity measurements were repeated three times to ensure the consistency and accuracy of the results. Mechanical properties were tested by a material testing system (XTD-500E; Jingkong Mechanical Testing Co., Ltd.). To measure the temperature during the RSMO, a K-type thermocouple (KAIPUSEN) was positioned on the surface of the sample (PVA/BN-Ga, size: $2.5 \text{ cm} \times 2.5 \text{ cm} \times 120 \text{ } \mu\text{m}$). Additionally, a pressure sensor (BD-24042802; Yubo Intelligence) was located at its center. The temperature and pressure data were obtained using a data acquisition instrument (34972A, Agilent) (Figure S29). The insulation of the TIM was simply verified using a multimeter (DT9205A, Milaike). The samples ($15 \text{ mm} \times 15 \text{ mm} \times 0.06 \text{ mm}$) were

placed between the LEDs and heat sinks, and the surface temperatures of the LED lamps were recorded using an infrared thermal imager (Fotric 220s). The output power, open-circuit voltage, and closed-circuit voltage of the thermoelectric devices were tested using an electronic load instrument (IT 8811A, ITECH).

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